Chapter 10. Applications of Theta Functions

Jacobi triple product, transformation laws, partitions, the two- and four-squares theorems

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Overview

The *theta function*

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}$$

converges for all $z \in \mathbb{C}$ and τ in the upper half-plane. It has a dual nature: as a function of z it belongs to the world of elliptic functions (period 1, "quasi-period" τ); as a function of τ it is modular and connects with partitions and with representing integers as sums of squares. The main tools are the *triple product formula* and the *transformation law*. We then discuss partitions and prove the celebrated two-squares and four-squares theorems.

1 Product formula for the Jacobi theta function

For $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$, **Jacobi's theta function** is

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}. \tag{1}$$

Special cases: $\theta(\tau) = \sum_n e^{\pi i n^2 \tau} = \Theta(0|\tau)$ for $\tau \in \mathbb{H}$, and $\vartheta(t) = \sum_n e^{-\pi n^2 t} = \theta(it)$ for $t > 0$. The heat kernel on the circle (Book I) is $H_t(x) = \sum_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = \Theta(x|4\pi i t)$; and the functional equation of ϑ gave that of ζ (Chapter 6).

Proposition 1.1

The function Θ satisfies the following properties:
(i) Θ is entire in $z \in \mathbb{C}$ and holomorphic in $\tau \in \mathbb{H}$.
(ii) $\Theta(z+1|\tau) = \Theta(z|\tau)$.
(iii) $\Theta(z+\tau|\tau) = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}$.
(iv) $\Theta(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$.

Proof. If $\mathrm{Im}(\tau) = t \geq t_0 > 0$ and $|z| \leq M$, then $\sum_n |e^{\pi i n^2 \tau} e^{2\pi i n z}| \leq C \sum_{n \geq 0} e^{-\pi n^2 t_0} e^{2\pi n M} < \infty$, uniformly; this gives (i).

(ii) is clear since $e^{2\pi i n z}$ has period 1. For (iii), complete the square:

$$\Theta(z+\tau|\tau) = \sum_n e^{\pi i (n^2+2n)\tau} e^{2\pi i n z} = \sum_n e^{\pi i (n+1)^2 \tau} e^{-\pi i \tau} e^{2\pi i (n+1) z} e^{-2\pi i z} = \Theta(z|\tau) e^{-\pi i \tau} e^{-2\pi i z}.$$

By (ii), (iii) it suffices for (iv) to show $\Theta(1/2 + \tau/2|\tau) = 0$: $\Theta(1/2+\tau/2|\tau) = \sum_n (-1)^n e^{\pi i (n^2+n)\tau}$, and pairing $n \geq 0$ with $-n-1$, which has opposite parity and $(-n-1)^2 + (-n-1) = n^2 + n$, the terms cancel. ∎

With the standard notation $q = e^{\pi i \tau}$, the **triple product** is

$$\Pi(z|\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} e^{2\pi i z})(1 + q^{2n-1} e^{-2\pi i z}).$$

Proposition 1.2

The function $\Pi(z|\tau)$ satisfies the following properties:
(i) $\Pi(z|\tau)$ is entire in $z \in \mathbb{C}$ and holomorphic for $\tau \in \mathbb{H}$.
(ii) $\Pi(z+1|\tau) = \Pi(z|\tau)$.

(iii) $\Pi(z+\tau|\tau) = \Pi(z|\tau)e^{-\pi i\tau}e^{-2\pi iz}$.
(iv) $\Pi(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$. Moreover, these points are simple zeros of $\Pi(\cdot|\tau)$, and $\Pi(\cdot|\tau)$ has no other zeros.

Proof. If $\mathrm{Im}(\tau) \geq t_0 > 0$, $|q| \leq e^{-\pi t_0} < 1$ and each factor is $1+O(|q|^{2n-1}e^{2\pi|z|})$; since $\sum|q|^{2n-1} < \infty$, the results on infinite products (Section 3 of Chapter 5) give (i).

(ii) is clear. For (iii), since $q^2 = e^{2\pi i\tau}$,

$$\Pi(z+\tau|\tau) = \prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n+1}e^{2\pi iz})(1+q^{2n-3}e^{-2\pi iz}),$$

and comparing with $\Pi(z|\tau)$, $\Pi(z+\tau|\tau) = \Pi(z|\tau)\frac{1+q^{-1}e^{-2\pi iz}}{1+qe^{2\pi iz}}$;

(iii) follows from $(1+x)/(1+x^{-1}) = x$ for $x \neq -1$. For (iv), a convergent product vanishes only if a factor does. $1-q^{2n} \neq 0$. $1+q^{2n-1}e^{2\pi iz} = 0$ iff $q^{2n-1}e^{2\pi iz} = -1 = e^{\pi i}$, i.e. $(2n-1)\tau + 2z \equiv 1 \pmod 2$, i.e. $z \equiv 1/2 + \tau/2 - n\tau \pmod 1$ $(n \geq 1)$. $1+q^{2n-1}e^{-2\pi iz} = 0$ iff $(2n-1)\tau - 2z \equiv 1 \pmod 2$, i.e. $z \equiv -1/2 - \tau/2 + n\tau \equiv 1/2 + \tau/2 + (n-1)\tau \pmod 1$ $(n \geq 1)$. This exhausts the zeros, and they are simple since $e^w - 1$ vanishes to order 1 at 0. ∎

Theorem 1.3 (Product formula)

For all $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ we have the identity $\Theta(z|\tau) = \Pi(z|\tau)$.

Proof. Fix $\tau \in \mathbb{H}$. There is a constant $c(\tau)$ with

$$\Theta(z|\tau) = c(\tau)\Pi(z|\tau)\,: \tag{2}$$

$F(z) = \Theta(z|\tau)/\Pi(z|\tau)$ is entire (by (iv) of both propositions) and doubly periodic with periods 1, τ (by (ii), (iii)), hence constant. We show $c(\tau) = c(4\tau)$. Put $z = 1/2$ in (2) ($e^{\pm 2\pi iz} = -1$):

$$\sum_n (-1)^n q^{n^2} = c(\tau)\prod_{n=1}^{\infty}(1-q^{2n})(1-q^{2n-1})^2 = c(\tau)\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1}), \quad c(\tau) = \frac{\sum_n(-1)^n q^{n^2}}{\prod_{n=1}^{\infty}(1-q^n)(1-q^{2n-1})}. \tag{3}$$

Put $z = 1/4$ ($e^{2\pi iz} = i$): $\Theta(1/4|\tau) = \sum_n q^{n^2}i^n$, and since $1/i = -i$, only even $n = 2m$ survive: $\Theta(1/4|\tau) = \sum_m q^{4m^2}(-1)^m$. On the other hand

$$\Pi(1/4|\tau) = \prod_{m=1}^{\infty}(1-q^{2m})(1+iq^{2m-1})(1-iq^{2m-1}) = \prod_{m=1}^{\infty}(1-q^{2m})(1+q^{4m-2}) = \prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4}),$$

the last step by splitting $2m = 4n$ and $2m = 4n-2$ in the first factor ($(1-q^{4n-2})(1+q^{4n-2}) = 1-q^{8n-4}$). Hence

$$c(\tau) = \frac{\sum_n(-1)^n q^{4n^2}}{\prod_{n=1}^{\infty}(1-q^{4n})(1-q^{8n-4})}, \tag{4}$$

and comparing (3) and (4) ($q \mapsto q^4$ corresponds to $\tau \mapsto 4\tau$), $c(\tau) = c(4\tau)$. So $c(\tau) = c(4^k\tau)$, and since $q^{4^k} = e^{\pi i 4^k\tau} \to 0$, (3) gives $c(4^k\tau) \to 1$; so $c(\tau) = 1$. ∎

Corollary 1.4

If $\mathrm{Im}(\tau) > 0$ and $q = e^{\pi i\tau}$, then

$$\theta(\tau) = \prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n-1})^2.$$

Thus $\theta(\tau) \neq 0$ for $\tau \in \mathbb{H}$.

Corollary 1.5

For each fixed $\tau \in \mathbb{H}$, the quotient

$$(\log \Theta(z|\tau))'' = \frac{\Theta(z|\tau)\Theta''(z|\tau) - (\Theta'(z|\tau))^2}{\Theta(z|\tau)^2}$$

is an elliptic function of order 2 with periods 1 and τ, and with a double pole at $z = 1/2+\tau/2$.

(Primes denote differentiation in z.)

Proof. Let $F = (\log \Theta)' = \Theta'/\Theta$. Differentiating (ii), (iii) of Proposition 1.1 gives $F(z+1) = F(z)$ and $F(z+\tau) = F(z) - 2\pi i$, so F' is doubly periodic. Θ vanishes in the fundamental parallelogram only at $1/2 + \tau/2$, simply, so F has only a simple pole there and F' only a double pole. ■

1.1 Further transformation laws

In Chapter 9 the modular character was expressed through $T_1 : \tau \mapsto \tau + 1$ and $S : \tau \mapsto -1/\tau$. For Θ it is natural to use $T_2 : \tau \mapsto \tau + 2$ and S, since $\Theta(z|\tau + 2) = \Theta(z|\tau)$ but $\Theta(z|\tau + 1) \neq \Theta(z|\tau)$.

Theorem 1.6

If $\tau \in \mathbb{H}$, then

$$\Theta(z|-1/\tau) = \sqrt{\tau/i}\, e^{\pi i z^2 \tau} \Theta(z\tau|\tau) \qquad \text{for all } z \in \mathbb{C}. \tag{5}$$

Here $\sqrt{\tau/i}$ denotes the branch of the square root defined on the upper half-plane that is positive when $\tau = it$, $t > 0$.

Proof. It suffices to prove (5) for $z = x$ real and $\tau = it$, $t > 0$: for fixed real x both sides are holomorphic in $\tau \in \mathbb{H}$ and would agree on the positive imaginary axis; and for fixed τ both sides are entire in z and would agree on $\mathbb{R}$. For $z = x$, $\tau = it$, (5) reads

$$\sum_n e^{-\pi n^2/t} e^{2\pi i n x} = t^{1/2} e^{-\pi t x^2} \sum_n e^{-\pi n^2 t} e^{-2\pi n x t}.$$

Replacing x by a, this is $\sum_n e^{-\pi t(n+a)^2} = \sum_n t^{-1/2} e^{-\pi n^2/t} e^{2\pi i n a}$, which is equation (3) of Chapter 4 (from Poisson summation). ■

Corollary 1.7

If $\mathrm{Im}(\tau) > 0$, then $\theta(-1/\tau) = \sqrt{\tau/i}\,\theta(\tau)$.

(Set $z = 0$; for $\tau = it$ this is the functional equation of ϑ.) This gives precise information about θ as $\tau \to 0$; the next corollary describes θ near $\tau = 1$.

Corollary 1.8

If $\tau \in \mathbb{H}$, then

$$\theta(1 - 1/\tau) = \sqrt{\tau/i} \sum_{n=-\infty}^{\infty} e^{\pi i (n+1/2)^2 \tau} = \sqrt{\tau/i}\,(2e^{\pi i \tau/4} + \cdots).$$

The second identity means that $\theta(1-1/\tau) \sim 2\sqrt{\tau/i}\, e^{\pi i\tau/4}$ as $\mathrm{Im}(\tau) \to \infty$.

Proof. n and n^2 have the same parity, so $\theta(1+\tau) = \sum_n (-1)^n e^{\pi i n^2 \tau} = \Theta(1/2|\tau)$; hence $\theta(1-1/\tau) = \Theta(1/2|-1/\tau)$. By Theorem 1.6 with $z = 1/2$,

$$\theta(1-1/\tau) = \sqrt{\tau/i}\, e^{\pi i\tau/4}\Theta(\tau/2|\tau) = \sqrt{\tau/i}\, e^{\pi i\tau/4} \sum_n e^{\pi i n^2\tau} e^{\pi i n\tau} = \sqrt{\tau/i} \sum_n e^{\pi i (n+1/2)^2 \tau}.$$

The terms $n = 0, -1$ contribute $2e^{\pi i\tau/4}$, of absolute value $2e^{-\pi t/4}$ ($\tau = \sigma + it$); the others total $O(\sum_{k\geq 1} e^{-\pi(k+1/2)^2 t}) = O(e^{-9\pi t/4})$. ∎

The **Dedekind eta function** is, for $\mathrm{Im}(\tau) > 0$,

$$\eta(\tau) = e^{\frac{\pi i\tau}{12}} \prod_{n=1}^{\infty} (1 - e^{2\pi i n\tau}).$$

Proposition 1.9

If $\mathrm{Im}(\tau) > 0$, then $\eta(-1/\tau) = \sqrt{\tau/i}\, \eta(\tau)$.

Proof. Differentiate (5) and evaluate at $z_0 = 1/2 + \tau/2$. By the product formula,

$$\Theta(z|\tau) = (1 + qe^{-2\pi iz}) \prod_{n=1}^{\infty} (1-q^{2n})(1+q^{2n-1}e^{2\pi iz})(1+q^{2n+1}e^{-2\pi iz}),$$

and the first factor vanishes at z_0 (where $e^{2\pi i z_0} = -q$), with derivative $-2\pi i q e^{-2\pi i z_0} = 2\pi i$ there, while the remaining factors equal $(1-q^{2n})^3$. So

$$\Theta'(z_0|\tau) = 2\pi i H(\tau), \qquad H(\tau) = \prod_{n=1}^{\infty} (1 - e^{2\pi i n\tau})^3.$$

Replacing $-1/\tau$ by τ in (5), $\Theta(z|\tau) = \sqrt{i/\tau}\, e^{-\pi i z^2/\tau}\Theta(-z/\tau|-1/\tau)$. Differentiating and evaluating at z_0 (where $-z_0/\tau = -\frac{1}{2\tau} - \frac{1}{2}$ is a zero of $\Theta(\cdot|-1/\tau)$, congruent to $\frac{1}{2} + \frac{-1/\tau}{2}$), a computation gives

$$2\pi i H(\tau) = \sqrt{i/\tau}\, e^{-\frac{\pi i\tau}{4}} e^{-\frac{\pi i}{2}} e^{-\frac{\pi i}{4\tau}} \Big(-\frac{1}{\tau}\Big) 2\pi i H(-1/\tau), \qquad \text{i.e.} \qquad e^{\frac{\pi i\tau}{4}} H(\tau) = \Big(\frac{i}{\tau}\Big)^{3/2} e^{-\frac{\pi i}{4\tau}} H(-1/\tau),$$

which says $\eta(\tau)^3 = (i/\tau)^{3/2}\eta(-1/\tau)^3$. For $\tau = it$, $\eta(\tau) > 0$, so taking cube roots gives $\eta(\tau) = \sqrt{i/\tau}\, \eta(-1/\tau)$; by analytic continuation this holds on $\mathbb{H}$. ∎

2 Generating functions

For a sequence $\{F_n\}_{n=0}^{\infty}$, its **generating function** is $F(x) = \sum_{n=0}^{\infty} F_n x^n$; properties of the sequence translate into algebraic or analytic properties of F, which can lead back to new insights. The **partition function** $p(n)$ is the number of ways n can be written as a sum of positive integers ($p(0) = 1$; e.g. $p(1) = 1$, $p(2) = 2$, $p(3) = 3$, $p(4) = 5$, $p(5) = 7$, $p(6) = 11$, $p(7) = 15$, $p(8) = 22$, $p(12) = 77$).

Theorem 2.1 (Euler)

If $|x| < 1$, then $\displaystyle\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}$.

Formally, write $\frac{1}{1-x^k} = \sum_{m=0}^{\infty} x^{km}$ and multiply out: grouping equal parts, a partition is $n = m_1k_1 + \cdots + m_rk_r$ with distinct k_j, corresponding to the term $(x^{k_1})^{m_1} \cdots (x^{k_r})^{m_r}$. The justification is as for the product formula for ζ (Section 1 of Chapter 7), using convergence of $\prod 1/(1-x^k)$, since $\frac{1}{1-x^k} = 1 + O(x^k)$.

Similarly $\prod 1/(1-x^{2n-1})$ generates $p_o(n)$, the number of partitions of n into odd parts, and $\prod(1+x^n)$ generates $p_u(n)$, partitions into unequal parts. Remarkably $p_o(n) = p_u(n)$, i.e.

$$\prod_{n=1}^{\infty} \frac{1}{1-x^{2n-1}} = \prod_{n=1}^{\infty} (1+x^n) :$$

indeed $\prod(1+x^n)\prod(1-x^n) = \prod(1-x^{2n})$, and by parity $\prod(1-x^{2n})\prod(1-x^{2n-1}) = \prod(1-x^n)$.

Let $p_{e,u}(n)$, $p_{o,u}(n)$ be the numbers of partitions of n into an even, resp. odd, number of unequal parts. Euler proved $p_{e,u}(n) = p_{o,u}(n)$ unless n is a **pentagonal number** $k(3k+1)/2$, $k \in \mathbb{Z}$ (namely $1, 2, 5, 7, 12, 15, 22, 26, \ldots$), in which case $p_{e,u}(n) - p_{o,u}(n) = (-1)^k$. Since multiplying out gives terms $(-1)^r x^{n_1+\cdots+n_r}$ with distinct n_j,

$$\prod_{n=1}^{\infty}(1-x^n) = \sum_{n=0}^{\infty}[p_{e,u}(n) - p_{o,u}(n)]x^n,$$

so Euler's theorem follows from:

Proposition 2.2 (Euler's pentagonal number theorem)

$$\prod_{n=1}^{\infty}(1-x^n) = \sum_{k=-\infty}^{\infty}(-1)^k x^{\frac{k(3k+1)}{2}}.$$

Proof. Set $x = e^{2\pi iu}$. In the triple product take $q = e^{3\pi iu}$ and $z = 1/2 + u/2$; then

$$\prod_{n=1}^{\infty}(1-e^{2\pi i3nu})(1-e^{2\pi i(3n-1)u})(1-e^{2\pi i(3n-2)u}) = \prod_{n=1}^{\infty}(1-e^{2\pi inu}).$$

By Theorem 1.3 the product equals $\sum_n e^{3\pi in^2u}(-1)^n e^{2\pi inu/2} = \sum_n (-1)^n e^{\pi in(3n+1)u} = \sum_n (-1)^n x^{n(3n+1)/2}$. ∎

The growth of $p(n)$ can be analyzed via $1/\prod(1-x^n)$ as $|x| \to 1$; elementary considerations give a rough order of growth, and the transformation law of η (Proposition 1.9) leads to a precise asymptotic formula (Appendix A).

3 The theorems about sums of squares

Let $r_2(n)$ be the number of pairs $(x, y) \in \mathbb{Z}^2$ with $n = x^2 + y^2$ (e.g. $r_2(3) = 0$, $r_2(5) = 8$). *Sum of two squares:* which integers are sums of two squares, and what is $r_2(n)$? No integer $4k+3$ is a sum of two squares, and no integer $8k+7$ is a sum of three squares. Let $r_4(n)$ be the number of representations of n as a sum of four squares. *Sum of four squares:* is every positive integer a sum of four squares, and what is $r_4(n)$? These problems, going back to the third century, were first fully solved with Jacobi's theta functions.

3.1 The two-squares theorem

If $n = a^2 + b^2$ and $m = c^2 + d^2$, then with $x + iy = (a + ib)(c + id)$, $x^2 + y^2 = nm$; so divisibility properties of n matter. Let $d_1(n)$ and $d_3(n)$ be the numbers of divisors of n of the form $4k+1$ and $4k+3$.

Theorem 3.1 (Two-squares theorem)

If $n \geq 1$, then $r_2(n) = 4(d_1(n) - d_3(n))$.

Consequently, if $n = p_1^{a_1} \cdots p_r^{a_r}$ with distinct primes p_j, then n is a sum of two squares iff every prime p_j of the form $4k+3$ has an even exponent a_j. The starting point: for $q = e^{\pi i \tau}$, $\tau \in \mathbb{H}$,

$$\theta(\tau)^2 = \sum_{n=0}^{\infty} r_2(n) q^n, \tag{6}$$

since $\theta(\tau)^2 = \sum_{n_1} \sum_{n_2} q^{n_1^2} q^{n_2^2} = \sum_{(n_1,n_2)\in\mathbb{Z}^2} q^{n_1^2+n_2^2}$.

Proposition 3.2

The identity $r_2(n) = 4(d_1(n) - d_3(n))$, $n \geq 1$, is equivalent to the identities

$$\theta(\tau)^2 = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = 1 + 4 \sum_{n=1}^{\infty} \frac{q^n}{1 + q^{2n}}, \tag{7}$$

whenever $q = e^{\pi i \tau}$ and $\tau \in \mathbb{H}$.

Proof. Both series converge absolutely ($|q| < 1$), and they are equal since $1/(q^n + q^{-n}) = q^{|n|}/(1 + q^{2|n|})$. Since $(1 + q^{2n})^{-1} = (1 - q^{2n})/(1 - q^{4n})$, the right side of (7) is $1 + 4\sum_{n\geq 1} \left[\frac{q^n}{1-q^{4n}} - \frac{q^{3n}}{1-q^{4n}}\right]$. Now $\frac{1}{1-q^{4n}} = \sum_{m\geq 0} q^{4nm}$, so

$$\sum_{n=1}^{\infty} \frac{q^n}{1-q^{4n}} = \sum_{n=1}^{\infty} \sum_{m=0}^{\infty} q^{n(4m+1)} = \sum_{k=1}^{\infty} d_1(k) q^k, \qquad \sum_{n=1}^{\infty} \frac{q^{3n}}{1-q^{4n}} = \sum_{k=1}^{\infty} d_3(k) q^k$$

(the series converge since $d_j(k) \leq k$). Compare with (6). ∎

Let

$$C(\tau) = 2 \sum_{n=-\infty}^{\infty} \frac{1}{q^n + q^{-n}} = \sum_{n=-\infty}^{\infty} \frac{1}{\cos(\pi n \tau)}, \qquad q = e^{\pi i \tau},\ \tau \in \mathbb{H}. \tag{8}$$

We must prove $\theta^2 = C$. There is a striking parallel: θ comes from the heat equation on the line, whose kernel is the Gaussian $e^{-\pi x^2}$, its own Fourier transform, and its transformation law comes from Poisson summation; C comes from the steady-state heat equation in a strip, whose kernel is $1/\cosh \pi x$ (Section 1), again its own Fourier transform (Example 3 in Section 2 of Chapter 3), and its transformation law again comes from Poisson summation. Indeed, $\theta(\tau)^2 = (i/\tau)\theta(-1/\tau)^2$ (Corollary 1.7); and (5) of Chapter 4 with $a = 0$ gives $\sum_n \frac{1}{\cosh \pi n t} = \frac{1}{t} \sum_n \frac{1}{\cosh(\pi n/t)}$, i.e. $C(\tau) = (i/\tau)C(-1/\tau)$ for $\tau = it$, hence for all $\tau \in \mathbb{H}$. Both θ^2 and C tend to 1 as $\mathrm{Im}(\tau) \to \infty$. At the "cusp" $\tau = 1$: by Corollary 1.8, $\theta(1 - 1/\tau)^2 \sim 4(\tau/i)e^{\pi i \tau/2}$; and (5) of Chapter 4 with $a = 1/2$ gives $\sum_n \frac{(-1)^n}{\cosh(\pi n/t)} = t \sum_n \frac{1}{\cosh(\pi(n+1/2)t)}$, so by analytic continuation

$$C(1 - 1/\tau) = \frac{\tau}{i} \sum_n \frac{1}{\cos(\pi(n + 1/2)\tau)} = 4\frac{\tau}{i} e^{\pi i \tau/2} + O(|\tau| e^{-3\pi t/2}) \quad (t \to \infty),$$

the main terms being $n = 0, -1$.

Proposition 3.3

The function $C(\tau) = \sum 1/\cos(\pi n\tau)$, defined in the upper half-plane, satisfies (i) $C(\tau+2) = C(\tau)$.
(ii) $C(\tau) = (i/\tau)C(-1/\tau)$.
(iii) $C(\tau) \to 1$ as $\mathrm{Im}(\tau) \to \infty$.
(iv) $C(1-1/\tau) \sim 4(\tau/i)e^{\pi i\tau/2}$ as $\mathrm{Im}(\tau) \to \infty$. Moreover, $\theta(\tau)^2$ satisfies the same properties.

Theorem 3.4

Suppose f is a holomorphic function in the upper half-plane that satisfies:
(i) $f(\tau+2) = f(\tau)$, (ii) $f(-1/\tau) = f(\tau)$, (iii) $f(\tau)$ is bounded, then f is constant.

Let $\mathcal{F} = \{\tau \in \overline{\mathbb{H}} : |\mathrm{Re}(\tau)| \le 1,\ |\tau| \ge 1\}$; the points $\tau = \pm 1$ are its **cusps** (equivalent under $\tau \mapsto \tau+2$).

Lemma 3.5

Every point in the upper half-plane can be mapped into $\mathcal{F}$ using repeatedly one or another of the following fractional linear transformations or their inverses: $T_2 : \tau \mapsto \tau+2$, $S : \tau \mapsto -1/\tau$.

$\mathcal{F}$ is the **fundamental domain** for the group G generated by T_2 and S (uniqueness of representatives fails only on $\partial\mathcal{F}$). Elements $g \in G$ are represented by matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $g(\tau) = \frac{a\tau+b}{c\tau+d}$; these have integer entries and determinant 1, so

$$\mathrm{Im}(g(\tau)) = \frac{\mathrm{Im}(\tau)}{|c\tau+d|^2}. \tag{9}$$

Proof. For $\tau \in \mathbb{H}$, since c, d are integers, by (9) we may choose $g_0 \in G$ with $\mathrm{Im}(g_0(\tau))$ maximal. Translations $T_2^{\pm 1}$ do not change imaginary parts, so there is $g_1 \in G$ with $|\mathrm{Re}(g_1(\tau))| \le 1$ and $\mathrm{Im}(g_1(\tau))$ maximal. If $|g_1(\tau)| < 1$, then $\mathrm{Im}(Sg_1(\tau)) = \mathrm{Im}(-1/g_1(\tau)) = \mathrm{Im}(g_1(\tau))/|g_1(\tau)|^2 > \mathrm{Im}(g_1(\tau))$, contradicting maximality. So $g_1(\tau) \in \mathcal{F}$. ∎

Proof of Theorem 3.4. Suppose f is not constant. Let $g(z) = f(\tau)$ with $z = e^{\pi i\tau}$; g is well defined in the punctured unit disc (period 2) and bounded near 0, so 0 is removable and $\lim_{\mathrm{Im}\,\tau\to\infty} f(\tau)$ exists; by the maximum modulus principle $\lim_{\mathrm{Im}\,\tau\to\infty}|f(\tau)| < \sup_{\mathcal{F}}|f|$. At the cusps (by $f(\tau+2) = f(\tau)$ it suffices to treat $\tau = 1$) we show $\lim_{\mathrm{Im}\,\tau\to\infty} f(1-1/\tau)$ exists and $\lim|f(1-1/\tau)| < \sup_{\mathcal{F}}|f|$, by interchanging 1 and ∞. Let U_n be the fractional linear transformation of $\begin{pmatrix} 1-n & n \\ -n & 1+n \end{pmatrix}$, which fixes 1; with $\nu(\tau) = 1/(1-\tau)$ (taking 1 to ∞, with inverse $\nu^{-1}(\tau) = 1-1/\tau$), $U_n = \nu^{-1}T_n\nu$, $T_n(\tau) = \tau+n$. Hence $U_nU_m = U_{n+m}$ and $U_{-1} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} = T_2S$. So every U_n is a finite composite of T_2, S and their inverses, and f is invariant under U_n: $f(\nu^{-1}T_n\nu(\tau)) = f(\tau)$. Thus $F(\tau) = f(\nu^{-1}(\tau)) = f(1-1/\tau)$ has period 1, and as above $h(z) = F(\tau)$, $z = e^{2\pi i\tau}$, has a removable singularity at 0, giving the claim by the maximum principle. Hence f attains its maximum in the interior of $\mathbb{H}$, contradicting the maximum principle. ∎

Proof of the two-squares theorem. $f(\tau) = C(\tau)/\theta(\tau)^2$ is holomorphic in $\mathbb{H}$ ($\theta \neq 0$ by Corollary 1.4) and invariant under T_2 and S (Proposition 3.3). In $\mathcal{F}$ it is bounded, tending to 1 as $\mathrm{Im}(\tau) \to \infty$ or $\tau \to \pm 1$, by (iii), (iv) of Proposition 3.3, verified by both C and θ^2. So f is bounded in $\mathbb{H}$, hence constant by Theorem 3.4, and the constant is 1: $\theta^2 = C$. ∎

3.2 The four-squares theorem

Let $\sigma_1^*(n)$ be the sum of the divisors of n that are not divisible by 4.

Theorem 3.6 (Four-squares theorem)

Every positive integer is the sum of four squares, and moreover $r_4(n) = 8\sigma_1^*(n)$ for all $n \geq 1$.

As before, $\theta(\tau)^4 = \sum_{n\geq 0} r_4(n)q^n$, $q = e^{\pi i\tau}$. Here nothing as simple as C is available; we need a subtle variant of the Eisenstein series. Recall the forbidden Eisenstein series $F(\tau) = \sum_m \sum_n \frac{1}{(n+m\tau)^2}$ (sum first in n, then m; $(n,m) \neq (0,0)$) of Corollary 2.6, and define

$$E_2^*(\tau) = F(\tau/2) - 4F(2\tau), \qquad \tau \in \mathbb{H}, \tag{10}$$

equivalently $E_2^*(\tau) = \sum_m \sum_n \left[\frac{1}{(n+m\tau/2)^2} - \frac{1}{(n/2+m\tau)^2}\right]$; the order of summation is critical.

Proposition 3.7

The assertion $r_4(n) = 8\sigma_1^*(n)$ is equivalent to the identity $\theta(\tau)^4 = -\pi^{-2}E_2^*(\tau)$, where $\tau \in \mathbb{H}$.

Proof. It suffices to show $-\pi^{-2}E_2^*(\tau) = 1 + 8\sum_{k\geq 1}\sigma_1^*(k)q^k$. By Corollary 2.6, $F(\tau) = \frac{\pi^2}{3} - 8\pi^2\sum_{k\geq 1}\sigma_1(k)e^{2\pi ik\tau}$, $\sigma_1(k)$ the sum of divisors of k. Now $\sigma_1^*(n) = \sigma_1(n)$ if $4 \nmid n$, and $\sigma_1^*(n) = \sigma_1(n) - 4\sigma_1(n/4)$ if $4 \mid n$ (a divisor $d = 4\tilde{d}$ of $n = 4\tilde{n}$ corresponds to the divisor $\tilde{d}$ of $\tilde{n}$). With (10),

$$E_2^*(\tau) = -\pi^2 - 8\pi^2\sum_{k=1}^{\infty}\sigma_1^*(k)e^{\pi ik\tau}.$$

■

So Theorem 3.6 reduces to $\theta^4 = -\pi^{-2}E_2^*$, and the key is that E_2^* has the same modular properties as θ^4.

Proposition 3.8

The function $E_2^*(\tau)$ defined in the upper half-plane has the following properties:
(i) $E_2^*(\tau+2) = E_2^*(\tau)$.
(ii) $E_2^*(\tau) = -\tau^{-2}E_2^*(-1/\tau)$.
(iii) $E_2^*(\tau) \to -\pi^2$ as $\mathrm{Im}(\tau) \to \infty$.
(iv) $|E_2^*(1-1/\tau)| = O(|\tau^2 e^{\pi i\tau}|)$ as $\mathrm{Im}(\tau) \to \infty$. Moreover $-\pi^2\theta^4$ has the same properties.

(i) is immediate from the definition. Consider F and its *reverse* $\tilde{F}(\tau) = \sum_n \sum_m \frac{1}{(n+m\tau)^2}$ (summing first in m).

Lemma 3.9

The functions F and $\tilde{F}$ satisfy:
(a) $F(-1/\tau) = \tau^2\tilde{F}(\tau)$, (b) $F(\tau) - \tilde{F}(\tau) = 2\pi i/\tau$, (c) $F(-1/\tau) = \tau^2 F(\tau) - 2\pi i\tau$.

Proof. (a) follows from the definitions and $(n+m(-1/\tau))^2 = \tau^{-2}(-m+n\tau)^2$. For (b) use $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$ with $\eta(\tau) = q^{1/12}\prod(1-q^{2n})$, $q = e^{\pi i\tau}$. By Proposition 3.2 of Chapter 5, the logarithmic derivative in τ is

$$\frac{\eta'}{\eta}(\tau) = \frac{\pi i}{12} - 2\pi i\sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}}, \qquad \sum_{n=1}^{\infty}\frac{nq^{2n}}{1-q^{2n}} = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} nq^{2nm} = \sum_{k=1}^{\infty}\sigma_1(k)q^{2k}.$$

Since $F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum_k \sigma_1(k) q^{2k}$, $\frac{\eta'}{\eta}(\tau) = \frac{i}{4\pi} F(\tau)$. By the chain rule the logarithmic derivative of $\eta(-1/\tau)$ is $\tau^{-2}\frac{\eta'}{\eta}(-1/\tau)$, which by (a) equals $\frac{i}{4\pi}\tilde{F}(\tau)$. Taking logarithmic derivatives in the functional equation for η,

$$\frac{i}{4\pi}\tilde{F}(\tau) = \frac{1}{2\tau} + \frac{i}{4\pi}F(\tau),$$

so $\tilde{F}(\tau) = F(\tau) - 2\pi i/\tau$.

(c) follows from (a) and (b). ∎

Proof of Proposition 3.8. (ii) By (10) and Lemma 3.9(c),

$$\begin{aligned} E_2^*(-1/\tau) &= F(-1/(2\tau)) - 4F(-2/\tau) = [4\tau^2 F(2\tau) - 4\pi i\tau] - 4[(\tau/2)^2 F(\tau/2) - \pi i\tau] \\ &= 4\tau^2 F(2\tau) - \tau^2 F(\tau/2) = -\tau^2(F(\tau/2) - 4F(2\tau)) = -\tau^2 E_2^*(\tau). \end{aligned}$$

(iii) In $F(\tau) = \frac{\pi^2}{3} - 8\pi^2 \sum \sigma_1(k) e^{2\pi i k\tau}$ the sum tends to 0 as $\operatorname{Im}(\tau) \to \infty$, so by (10) $E_2^*(\tau) \to \frac{\pi^2}{3} - 4\frac{\pi^2}{3} = -\pi^2$.

(iv) We first show

$$E_2^*(1 - 1/\tau) = \tau^2\Big[F\Big(\frac{\tau - 1}{2}\Big) - F(\tau/2)\Big]. \tag{11}$$

F has period 1, and Lemma 3.9(c) can be written $F(w) = w^{-2}[F(-1/w) + 2\pi i w]$. Let $u = (\tau - 1)/2$ and $w = u/\tau = \frac{1}{2} - \frac{1}{2\tau}$. Then $-1/w = -2 - \frac{2}{\tau - 1}$, so $F(-1/w) = F(-1/u) = u^2 F(u) - 2\pi i u$, and

$$F\Big(\frac{1}{2} - \frac{1}{2\tau}\Big) = \frac{\tau^2}{u^2}[u^2 F(u) - 2\pi i u + 2\pi i u/\tau] = \tau^2 F(u) - \frac{2\pi i \tau^2(1 - 1/\tau)}{u} = \tau^2 F\Big(\frac{\tau - 1}{2}\Big) - 4\pi i\tau.$$

Also $4F(2 - 2/\tau) = 4F(-1/(\tau/2)) = 4[(\tau/2)^2 F(\tau/2) - 2\pi i\tau/2] = \tau^2 F(\tau/2) - 4\pi i\tau$. Subtracting gives (11), since $E_2^*(1 - 1/\tau) = F(\frac{1}{2} - \frac{1}{2\tau}) - 4F(2 - \frac{2}{\tau})$. By the expansion of F, $F(\frac{\tau - 1}{2}) - F(\tau/2) = -8\pi^2 \sum_k \sigma_1(k) e^{\pi i k\tau}((-1)^k - 1) = 16\pi^2 e^{\pi i\tau} + O(|e^{3\pi i\tau}|)$, which with (11) gives (iv). For θ^4: (i)–(iii) follow from Section 1 ($\theta(-1/\tau)^4 = (\tau/i)^2\theta(\tau)^4 = -\tau^2\theta(\tau)^4$, $\theta \to 1$), and (iv) from Corollary 1.8: $\theta(1 - 1/\tau)^4 \sim 16(\tau/i)^2 e^{\pi i\tau}$. ∎

Proof of the four-squares theorem. Consider $f(\tau) = E_2^*(\tau)/\theta(\tau)^4$, holomorphic in $\mathbb{H}$, invariant under T_2 and S by Proposition 3.8. Since $\theta(\tau)^4 \to 1$ and $\theta(1 - 1/\tau)^4 \sim 16\tau^2 e^{\pi i\tau}(-1)$ as $\operatorname{Im}(\tau) \to \infty$, f is bounded in $\mathcal{F}$, hence in $\mathbb{H}$. By Theorem 3.4, f is constant, equal to $-\pi^2$ by (iii). So $\theta^4 = -\pi^{-2} E_2^*$, which proves the theorem; in particular $r_4(n) \geq 8 > 0$ for all $n \geq 1$. ∎